

Cultivated wild ginseng extracts upregulate the anti-apoptosis systems in cells and mice induced by bisphenol A

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Abstract Cultivated wild ginseng has a variety of pharmacological effects. The aim of this study was to investigate the effects of cultivated wild ginseng extracts (CWGE) on apoptosis. CWGE showed the ability to protect hormone production (testosterone and progesterone from *in vitro* results) in TM3 Leydig and TM4 Sertoli cells damaged by Bisphenol A (BPA). CWGE also showed the ability to protect production of testosterone (7.24%, $P < 0.05$), plasma luteinizing hormone (LH) (32.67%, $P < 0.05$), and follicle-stimulating hormone (FSH) (37.34%, $P < 0.05$) compared to the control and BPA-only groups. Reduction of apoptotic protein expression and induction of anti-apoptotic protein expression was observed in cells and rats pretreated with CWGE. These proteins are expressed through the ERK and p38 signalling pathways. In addition, CWGE might be a useful herbal medicine of cellular defense agents as our results in cells and animals.

Keywords: Cultivated wild ginseng extract, Apoptosis, Bisphenol A, Testicular toxicity

World Health Organization, if pregnancy is not possible, was defined as infertility including the cases that are not pregnancy during the two years without contraception¹. Reports suggest that approximately 15% of couples worldwide suffer from infertility caused by a wide range of physical and emotional factors. Infertility may be due to problems in the woman (40%) or man (40%), or due to unexplained causes (20%)^{2–5}. Because

75% of male infertility cases can be diagnosed easily but attract less social interest compared to infertility in females, more research is required to identify their root causes and possible solutions. Generally, male infertility is due to a decrease in sperm count and sperm motility. A number of endocrine-disrupting compounds such as BPA are found in the environment as an adverse result of industrial and manufacturing activity¹. BPA [2,2-bis (hydroxyphenyl) propane], one of the most widely produced chemicals in the world, is widely used to make a such as packaging materials and dental materials⁶. Based on the research results, BPA is in many exposed to humans in daily life⁷. Exposure of BPA was also confirmed to be a causative agent of reproductive dysfunction, both *in vitro* and *in vivo* including the reduced number of sperm in mice^{8–10}, structurally deformed destruction of the sperm and the apical ectoplasmic specialization^{11,12}, the collapse of the Blood-testis barrier¹³, and a reduction of the insertion frequency and performance¹⁴. It is therefore necessary to explore new preventative and therapeutic measures to address BPA-induced reproductive toxicity.

Germ cell death processes in testes have been reported to coincide with characteristics of apoptosis (programmed cell death)^{15,16}. Apoptosis has an important role in maintaining biological balance in cellular self-destruction¹⁷. Apoptosis of testicular germ cells affects spermatogenesis and causes abnormal spermatogenesis^{18,19}. Caspase-8, which is linked to cell death receptor and Fas receptor, which is the family of tumor necrosis death receptors were associated with cell apoptosis²⁰. The Bcl-2 family consists of anti-apoptotic genes such as Bcl-2 and Bcl-xL and apoptotic genes such as Bax, Bad, and Bim. Caspase-3 and caspase-9 are activated by cytochrome C and induce cell apoptosis²¹. Phosphorylation of protein kinase activates apoptotic factors (Bax) and caspase-2, and induces

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apoptosis. However, phosphorylation of protein kinase (stimulated by growth factor receptors) and cytokines activate survival factors (Bcl-2), inhibiting apoptosis.

Ginseng is widely used globally as an herbal dietary material. Ginsenosides extracted from ginseng may be also used as a therapeutic agent for several diseases^{22,23}. Ginsenosides found in ginseng are mainly divided into the protopanaxadiol (PPD) group having Rb1, Rc, Rd and Rg3 and protopanaxatriol (PPT) group having Re, Rf, Rg1 and Rh1^{24,25}.

Using cell and animal models, we studied the effects of CWGE on testicular anti-apoptosis, as well as its underlying molecular mechanisms.

Results & Discussion

In vitro hormone analysis

Figure 1(A) shows that production of testosterone was inhibited by BPA treatment (100 μ M) in TM3 cells. Nakamura *et al.*²⁶ reported that BPA directly affects the TM3 Leydig cells and decreases plasma and testicular testosterone levels. Testosterone levels tended to be significantly higher with treatment with 25 μ g/mL CWGE 25 and 100 μ M BPA compared to the treatment with 100 μ M BPA only (Figure 1(B)). Production of progesterone in TM3 and TM4 cells was significantly inhibited by BPA compared to the control (Figure 1(B)). TM3 and TM4 cells treated with CWGE and BPA showed an increase in production of progesterone compared to TM3 and TM4 cells treated with BPA alone (Figure 1(B)). Peretz and Flaws²⁷ reported that exposure to BPA inhibits the steroidogenic enzyme cytochrome P450 side-chain cleavage (*Cyp11a1*) and steroidogenic acute regulatory protein (*StAR*), leading to decreased progesterone and testosterone production. CWGE may be caused by the restoration of *Cyp11a1* and *StAR* gene expression²⁷. LH secreted from the pituitary can be involved in the synthesis of testosterone. In the Nakamura *et al.*²⁶ study, the high dose of BPA also significantly decreased plasma LH levels. Our result of decreased testosterone with high doses of BPA is similar. CWGE may reduce the adverse effects of BPA on the expression of cholesterol carrier protein or steroidogenic enzymes in the pituitary gland²⁶.

Testicular apoptotic and anti-apoptotic protein expression in BPA-treated TM3 and TM4 cells

To identify the effects of CWGE on BPA-induced apoptosis, TM3 and TM4 cells, TM3 and TM4 cells were treated with control medium or 100 μ M BPA for 24 hours or pre-treated with 25 μ g/mL CWGE for 1 h prior to BPA exposure. As shown in Figure 2(A), Fas

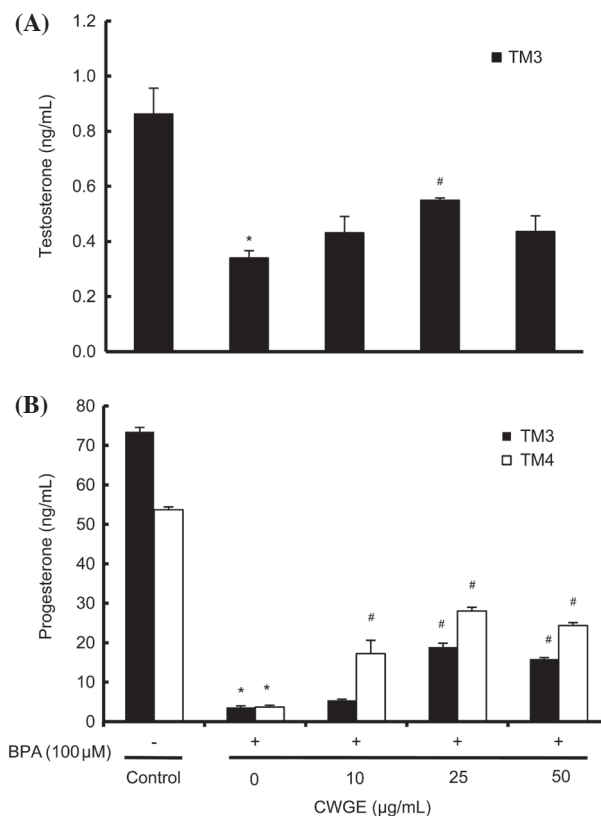


Figure 1. Effects of CWGE on testosterone and progesterone in BPA-treated TM3 (A) and TM4 (B) cells. Data have been presented as the means $\pm$ S.D. of three independent experiments. *#Significantly differed from the no treatment and BPA-treated control, respectively ($P < 0.05$).

protein expression was significantly increased when TM3 was treated with 100 μ M BPA, and reduced when pre-treated with 10 μ g/mL and 50 μ g/mL CWGE or 100 μ M N-acetyl-L-cysteine (NAC). Results showed that expression of apoptotic proteins (Fas, Bax, and Caspase-3) was significantly reduced in TM3 cells when they were pre-treated with 10 μ g/mL CWGE and BPA, while expression of the anti-apoptotic protein (Bcl-2) was significantly activated compared to BPA-only treatment (Figure 2(A)). An earlier study showed that BPA-treated germ cells undergo apoptosis through the Fas/FasL apoptotic pathway²⁸. The expression levels of anti-apoptotic and apoptotic proteins in TM4 cells treated with BPA were similar to those in TM3 cells treated with BPA. When TM4 cells were pre-treated with 25 μ g/mL CWGE and BPA, expression of apoptotic proteins was inhibited more than in other treatments and Bcl-2 expression was activated more than in other treatments (Figure 2(B)). Caspase-3 is importantly affected by downstream activity in the apoptotic pathway²⁹. Fas-mediated apoptosis is due to

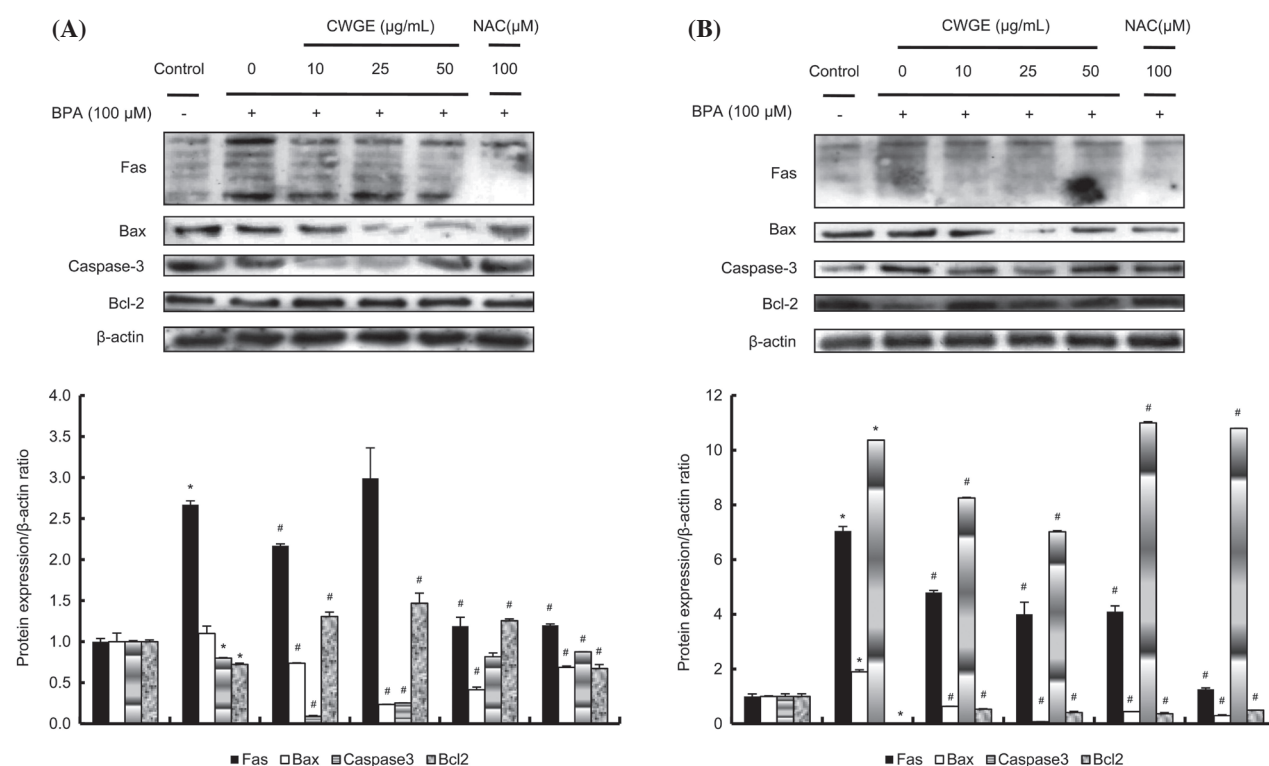


Figure 2. Effects of CWGE on anti-apoptosis and apoptosis protein expression in BPA-treated (A) TM3 and (B) TM4 cells. The ratio of protein expression between the Fas/Bax/Caspase-3/Bcl-2 and β -actin calculated. Each bar is presented as the means $\pm$ S.D. of three independent experiments. *.#Significantly different from the no treatment and BPA-treated control, respectively ($P < 0.05$).

the activity of caspase-8 in the death receptor and the mitochondrial pathways, and the activity of Bax in the mitochondrial pathway^{30,31}. Over expression of Bcl-2 such as anti-apoptotic protein protects an apoptosis of the cells³². In the present study, we found that the levels of Fas, Bax, Caspase-3, and Bcl-2 in TM3 and TM4 cells were changed by CWGE. According to these results, CWGE play a role in the prevention of apoptotic cell death in TM3 and TM4 cells. A recent study also demonstrated that Rg1 (a ginsenoside found in CWGE) is a notable anti-apoptotic molecule that is capable of blocking the caspase-dependent signalling cascade in Jurkat cells³³, Schwann cells³³, and PC 12 cells³⁴.

Phosphorylation of upstream kinases in BPA-treated TM3 and TM4 cells

Wang *et al.*³⁵ have indicated that ERK1/2, JNK and p38 (which are all mitogen-activated protein kinases, or MAPKs) are importantly affected by apoptosis. To investigate whether the protective effect of CWGE is fixed in relation to MAPK signalling, it was confirmed that ERK1/2 and p38 were activated in TM3 and TM4

cells by BPA stimulation³⁶. Sertoli cells are importantly affected the germ cell survival of the seminiferous epithelium and spermatogenesis³⁷. In addition, the protection of the TM3 and TM4 cells from toxic materials is critically important to spermatogenic function. To further elucidate the modulatory effects of CWGE on MAPK signalling pathways and their involvement in apoptotic and anti-apoptotic protein expression, cells were treated with specific inhibitors of each MAPK (U0126, an inhibitor of MEK1/2, and SB202190, an inhibitor of p38). The expression of Bax by treatment with CWGE and BPA was down-regulated in TM3 and TM4 cells treated with the p-ERK and p38-specific inhibitor (Figure 3). In other words, the ERK1/2 and p38 inhibitor also decreased Bax protein expression and increased expression of the Bcl-2 protein (an anti-apoptotic factor) in TM3 and TM4 cells. In previous study, Wang *et al.*³⁵ identified that ginsenosides have the effects of ERK1/2 phosphorylation and Bcl-2/Bax protein expression. This means that CWGE affects Bax and Bcl-2 expression through ERK1/2 and p38 pathways in TM3 and TM4 cells. These results show that male germ cell function is affected by MAPK signalling pathways and phosphorylation^{38,39}.

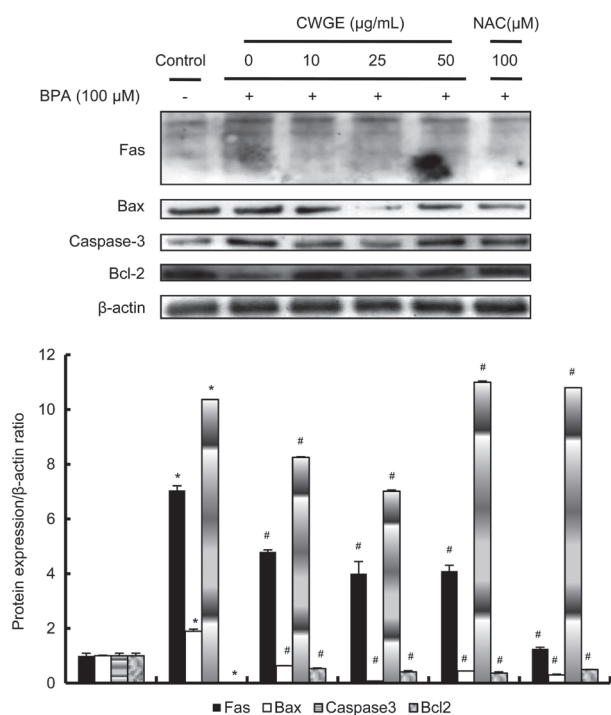


Figure 3. Effects of CWGE on anti-apoptosis and apoptosis protein expression in BPA-treated TM4 cells. The ratio of protein expression between the Fas/Bax/Caspase-3/Bcl-2 and β -actin was calculated. Each bar has been presented as the means $\pm$ S.D. of three independent experiments. *#Significantly differed from the 'no treatment' and BPA-treated control, respectively ($P < 0.05$).

In vivo hormone levels

Testosterone is important for male reproductive system. Testosterone is essential for male accessory sex gland function and structure maintenance. Insufficient testosterone may be caused the problem of spermatogenesis⁴⁰. The effects of CWGE on hormone levels in male rats treated with BPA were measured (Figure 4). The serum LH levels decreased by 42.01% ($P < 0.05$) in rats exposed to 25 mg BPA/(kg·d) (Group B) compared to Group A. These decreases in LH were inhibited by CWGE in Group D (32.67%, $P < 0.05$) and NAC (37.34%, $P < 0.05$) in Group C compared to Group A (Figure 4(A)). In addition, serum FSH levels in Group B (46.67%, $P < 0.05$) were significantly lower than in Group A and serum FSH levels in Group D (37.34%, $P < 0.05$) and Group C (28.01%, $P < 0.05$) were significantly higher than in Group B (Figure 4(B)). Serum testosterone was significantly decreased by 72.11% ($P < 0.05$) in Group B compared to Group A. However, testosterone in group D showed a 7.24% increase compared to group A (Figure 4(C)). We found that serum LH, FSH and testosterone tended to be lower in the BPA group

than in the control group, while CWGE appeared to prevent the BPA-induced decrease in LH, FSH, and testosterone levels. Testosterone levels *in vivo* (Figure 4) were similar to levels *in vitro* (Figure 1). These results may be that BPA disrupts the hypothalamic-pituitary-gonadal axis resulting in the relative gene expression (gonadotropin releasing hormone receptor (*Gnrhr*), luteinizing hormone beta (*Lhb*), follicle stimulating hormone beta (*Fshb*), estrogen receptor beta (*Esr2*) and androgen receptor (*Ar*), and estrogen receptor alpha (*Esr1*) with the changed testosterone, LH and FSH levels⁴¹. These results also supported findings that consumption of Panax ginseng Meyer extract is associated with increased testosterone levels in patients with fertility problems⁴².

Testicular apoptotic and anti-apoptotic protein expression in BPA-induced rats

The expression levels of the testicular apoptotic and anti-apoptotic proteins in rats after exposure to CWGE and BPA were evaluated by western blot analysis of the testes. As shown in Figure 5, injection of BPA (25 mg/kg) in Group B resulted in significant up-regulation of protein expression levels of Fas, Bax, and Caspase-3 and down-regulation of Bcl-2 protein expression compared with the normal control rats (Group A) (Figure 5). However, expression of all apoptotic proteins was reduced and expression of the Bcl-2 protein was increased in the 50 mg/kg CWGE group (Group D). Administration of 100 mg/kg NAC (Group C) also significantly increased expression of Bcl-2 and reduced expression of Bax, Caspase-3, and Fas in the testes compared to expression levels in the BPA group (Group B) (Figure 5). Similar to the *in vitro* results, these results suggest that CWGE-induced cleavage of Fas, Bax, and Caspase-3 leads to activation of the protease. Previous reports have demonstrated that Bcl-2 is the role of anti-apoptosis to protect the action of caspase protein^{43,44}. Expression of the Bcl-2 protein in Group D was increased, suggesting that CWGE-induced apoptosis might be associated with the activation of the caspase-like protease via a Bcl-2-insensitive signalling pathway. Therefore, CWGE seems to both induce and inhibit the activity and expression of testicular anti-apoptotic and apoptotic proteins *in vivo* as well as *in vitro*.

Terminal dUTP nick-end labelling (TUNEL) assay and sperm cell counts

Germ cell apoptosis was detected with a TUNEL assay. TUNEL-positive cells were brown, round, and showed DNA breaks in the seminiferous tubules⁴⁵. No apoptotic cells were found in the seminiferous tubules of the

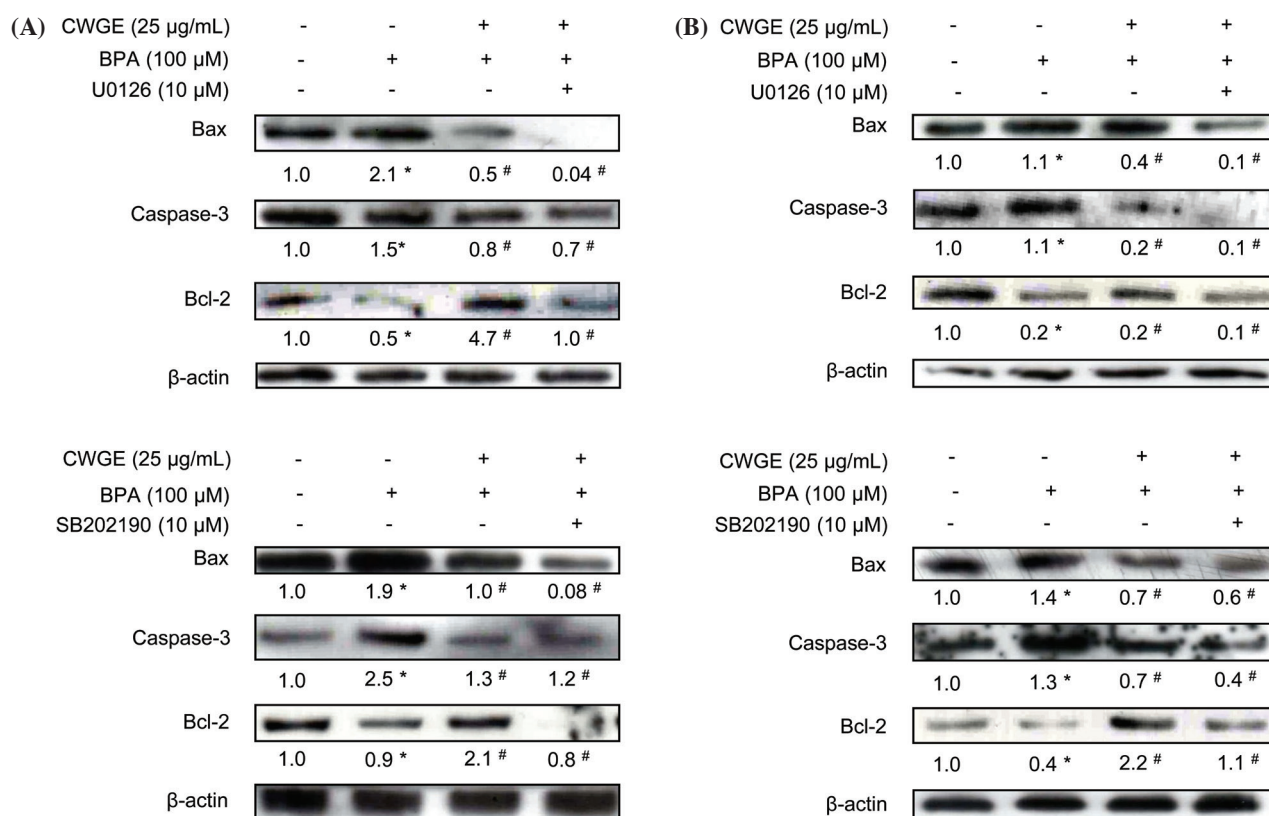


Figure 4. Effect of CWGE on upstream signaling pathways through MAP kinases of (A) TM3 and (B) TM4 cells. Data have been presented as the means $\pm$ S.D. of three independent experiments. The ratio of protein expression between the target protein and β -actin was calculated. */#Significantly differed from the ‘no treatment’ and BPA-treated control, respectively ($P < 0.05$).

control (Group A) rats (Figure 6(A)). There was a relatively low number of TUNEL-positive cells found using light microscopy in this study. This result may be due to the cause of phagocytosis^{46,47}. TUNEL-positive apoptotic spermatogonia and spermatocytes were observed in the seminiferous tubules of BPA-treated rats (Group B) (Figure 6(A)). Pre-treatment with CWGE (Group D) and NAC (Group C) resulted in a significant decrease of apoptotic germ cells compared treatment with BPA alone (Group B) (Figure 6(B)).

The effect of groups on the epididymal sperm count in male rats is shown in Figure 7C. The average number of sperm in rats treated with 25 mg/kg of BPA (Group B) compared to the control group (Group A) was significantly reduced. These results are consistent with the results of the reduction in the number of sperm in mice and BPA oral administration study^{48,49}. However, the increment of sperm cells was presented in pre-treatment with CWGE (240.60%, $P < 0.05$) and NAC (134.26%, $P < 0.05$) group compared to the only BPA group. We focused on the molecular mechanism of apoptosis in testicular cells (Leydig and Sertoli cells)

and in animal models. From this study, it can be concluded that CWGE has positive effects on infertility. Taken together, these results support that CWGE possess anti-apoptotic capabilities and suggest that CWGE might be a useful herbal medicine of cellular defense agents as our results in cells and animals.

Materials & Methods

Chemicals and reagents

CWGE was extracted using ethanol, as previously described by Ok *et al.*⁵⁰. The CWGE contained 6 main ginsenosides (Rg1, Rf, Re, Rh1, Rb1, and Rd), as determined HPLC analysis. Dulbecco’s modified Eagle Medium (DMEM), foetal bovine serum (FBS), and glutamine were obtained from HyClone (Logan, UT, USA). NAC and BPA were purchased from Sigma-Aldrich (St. Louis, MO, USA). All other chemicals were of the highest commercially available grade. β -actin polyclonal antibodies and peroxidase-conjugated

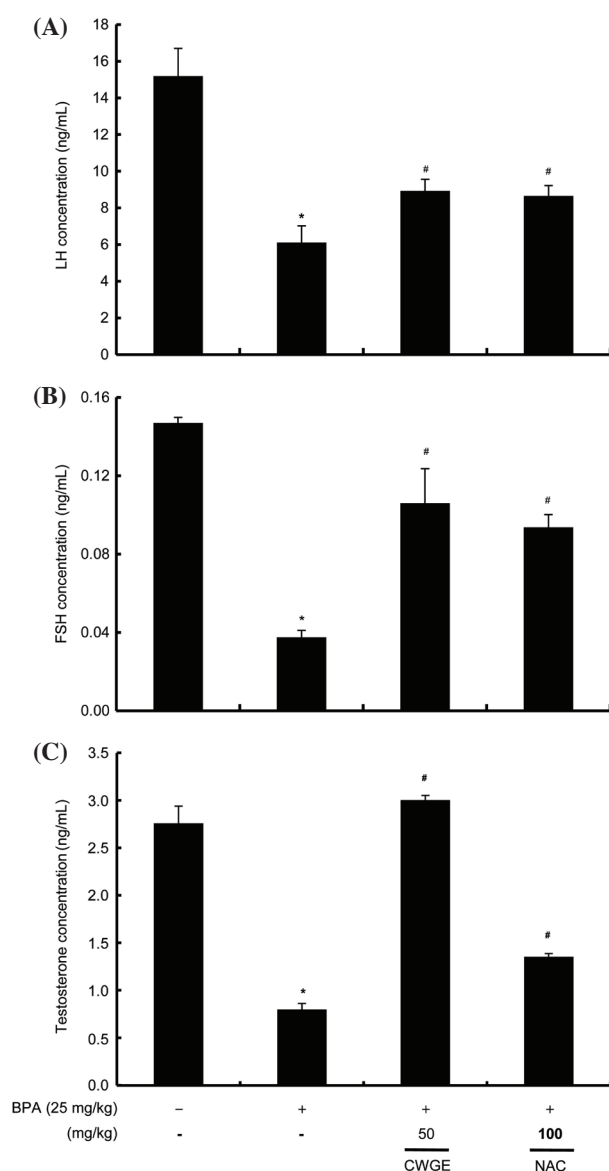


Figure 5. Effects of CWGE on LH, FSH and testosterone concentration in BPA-treated rats. Data have been presented as the means $\pm$ S.D. for 5 rats. *#Significantly differed from the ‘no treatment’ and BPA-treated control, respectively ($P < 0.05$).

anti-rabbit and anti-goat antibodies were purchased from Santa Cruz Biotechnology (Delaware, CA, USA). Fas/FasL, Caspase-3, Bax, Bcl2, Phospho-ERK1/2, phospho-JNK, phospho-p38, phospho-Akt polyclonal antibodies, ERK (U0126, an inhibitor of MEK1/2), and p38 (SB202190, an inhibitor of p38) were purchased from Cell Signaling Technology (Danvers, MA, USA).

Cell culture

The TM3 Leydig and TM4 Sertoli cell lines were

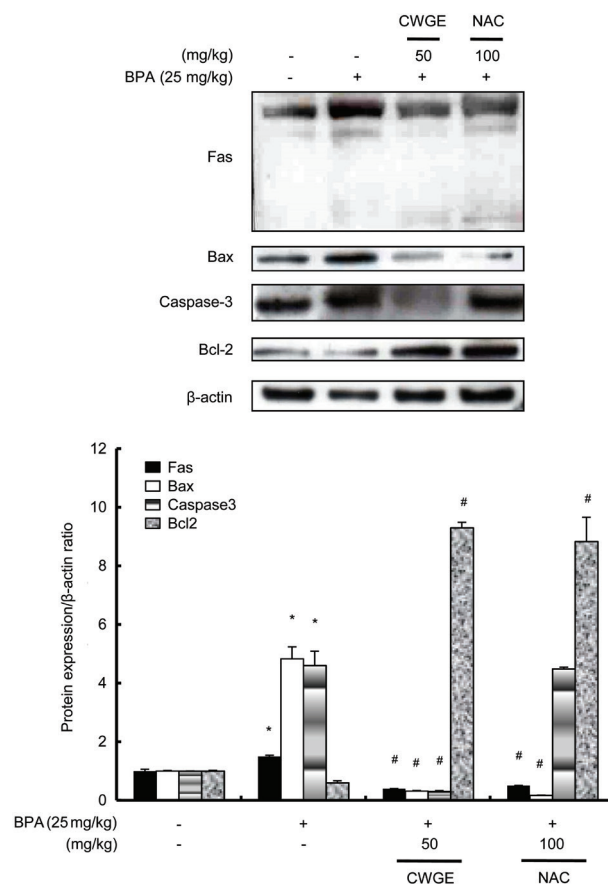


Figure 6. CWGE administration induces Bcl-2 and inhibits Fas, Bax and Caspase-3 protein expression in BPA-treated rats. Data have been presented as the means $\pm$ S.D. for 5 rats. *#Significantly differed from the ‘no treatment’ and BPA-treated control, respectively ($P < 0.05$).

purchased from the Korean Cell Line Bank (KCLB) (Seoul, Korea). TM3 and TM4 cells were cultured at 37°C and 5% CO₂ in DMEM containing 10% FBS, 100 units/mL penicillin, and 100 µg/mL streptomycin from HyClone Laboratories, Inc. (Logan, UT).

Preparation of whole-cell extracts

Whole-cell extracts were prepared according to the manufacturer’s instructions using cell lysis buffer containing 20 mM Tris (pH 7.5), 135 mM sodium chloride, 2 mM ethylenediaminetetraacetic acid (EDTA), 2 mM dithiothreitol, 25 mM β-clycerophosphate, 2 mM sodium pyrophosphate, 10% glycerol, 1% Triton X-100, 1 mM sodium orthovanadate, 10 mM NaF, 10 µg/mL aprotinin, 10 µg/mL leupeptin, and 1 mM phenylmethylsulfonyl fluoride. Protein concentrations were determined by the bicinchoninic acid (BCA) developed by Pierce Biotechnology (Rockford, IL, USA).

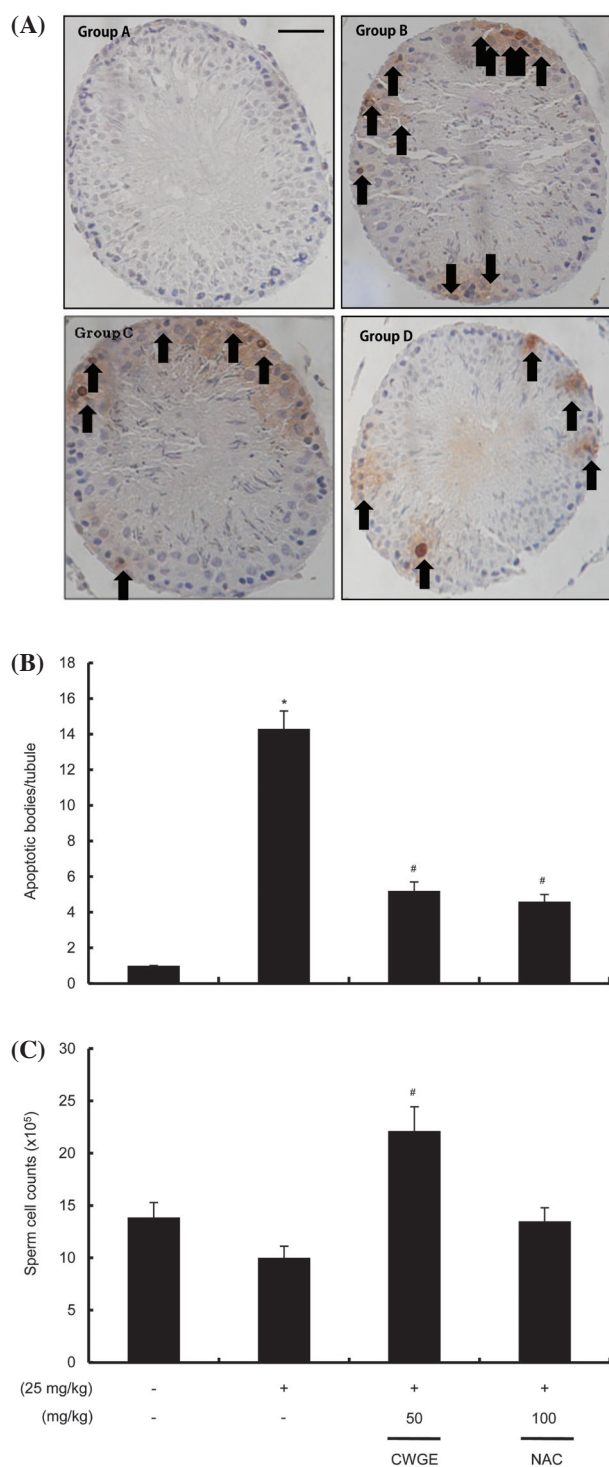


Figure 7. Terminal dUTP nick-end labeling (TUNEL) assay. (A) TUNEL staining of seminiferous tubule in rat testes, TUNEL-positive cells were brownish with a round shape (arrows), (B) The number of apoptotic (TUNEL-positive) cells per seminiferous tubule, (C) The number of sperm per rat in each group. Data have been presented as the means $\pm$ S.D. for 5 rats. *#Significantly differed from the 'no treatment' and BPA-treated control, respectively ($P < 0.05$).

Hormone levels assay

In the progesterone and testosterone assay, TM3 and TM4 cells were seeded in a six-well plate until they reached 80–90% confluency. After starvation for 12 hours, TM3 and TM4 cells were pre-treated with CWGE (10, 25 and 50 $\mu\text{g/mL}$) for 1 hour, treated with or without BPA (100 μM) for 24 hours, then switched to serum-free DMEM medium containing pregnenolone (25 μM) for 2 hours. The test chemical's vehicle was DMSO and its final concentration was 0.1%. Pregnenolone (3β -hydroxypregn-5-en-20-one) is an endogenous steroid hormone involved in the steroidogenesis of progestogens, mineralocorticoids, glucocorticoids, androgens, oestrogens, and neuroactive steroids. It is a prohormone and can be converted to progesterone. After treatment, the medium was collected and stored at -80°C for subsequent radioimmunoassay (RIA) analysis. Progesterone, testosterone, LH, and FSH were detected by an ELISA kit (Cayman systems) used according to the manufacturer's instructions.

Western blot analysis

TM3 and TM4 cells treated with various concentrations of CWGE, NAC, and BPA were washed with 1X-ice cold lysis buffer (pH 7.4). Cells were then harvested and lysed in 1X whole cell lysis buffer. Supernatants were collected as whole-cell lysates by centrifugation at 13,000 rpm for 15 min. The protein concentration was measured by the BCA protein assay kit (Pierce, Rockford, IL, USA). Proteins (20–30 $\mu\text{g/lane}$) were preheated at 95°C for 5 min. The samples were separated using 10% SDS-polyacrylamide gel electrophoresis (PAGE) and transferred onto a polyvinylidene difluoride (PVDF) membrane with a semidry transfer system from Bio-Rad (Hercules, CA, USA). The PVDF membrane was blocked in 5% non-fat milk solution for 3 hours at room temperature, and then incubated overnight at 4°C with primary antibody. After hybridization with primary antibody, the membrane was washed four times with phosphate-buffered saline containing 0.1% Tween 20 (PBST) for 5 min, incubated with horseradish peroxidase-conjugated secondary antibody for 1 hour at room temperature, and washed five times with PBST for 5 min. Final detection was performed with enhanced chemiluminescence (ECL) western blotting reagents from Santa Cruz Biotechnology Inc. (Delaware, CA, USA).

To gain quantitative insight into the effects of CWGE on BPA-induced apoptosis, TM3 and TM4 cells were pretreated with either CWGE (10, 25 and 50 $\mu\text{g/mL}$) or NAC (100 μM) for 1 hour, then treated with BPA (100 μM) for 24 hours. The test chemical's vehicle was DMSO and its final concentration was 0.1%. Then,

target antibody proteins (Fas, Bax, Caspase-3, Bcl-2, and β -actin) were detected by western blotting.

While investigating whether the protective effect of CWGE is related to MAPK signalling, it was confirmed that U0126 and SB202190 were activated after BPA stimulation of TM3 and TM4 cells. TM3 and TM4 cells were pretreated with 10 μ M of U0126 or SB202190 inhibitor for 1 hour, and then treated with 25 μ g/mL of CWGE for 1 hour. The phosphorylation of target antibody proteins (Fas, Bax, Caspase-3, Bcl-2 and β -actin) was detected by western blotting. All western blot bands were quantified using Image J program and then corrected by β -actin expression.

Animal experimental design

Twenty 7-week-old Sprague-Dawley rats were obtained from Samtako Biokorea (Kyongki, Korea). They were maintained under a 12 hours light/12 hours dark cycle under controlled temperature ($20 \pm 4^\circ\text{C}$), and relative humidity ($55 \pm 10\%$). All rats received humane care and unlimited access to food and water during the study. The rats were habituated to these conditions for one week before experiments were initiated. The rats were divided into four groups ($n=5$ per group). The rats were treated with BPA and CWGE at a daily dose of 50 mg/kg body weight. BPA, CWGE, and NAC were dissolved in ethanol, and then diluted with corn oil and saline. The control and test chemical's vehicle were ethanol and saline in corn oil and its final concentration was 10%. Control rats were treated with equal volumes of corn oil and saline (Group A). The BPA group, as a negative control, received 1 mL saline orally once per day for 1 week, then injected intraperitoneally with 1 mL BPA (50 mg/kg) dissolved in corn oil once a week (Group B) according to the BPA concentration of 50 mg/kg established by the U.S. Environmental Protection Agency (EPA) (U. S. EPA 1993)⁵¹. The NAC group, which served as a positive control, received an oral dose of 100 mg/kg of NAC (1 mL) once a day for one week, and then injected intraperitoneally with 50 mg/kg of BPA (1 mL) once a week (Group C). The CWGE group received 50 mg/kg of CWGE orally (1 mL) once a day for one week and then injected intraperitoneally with 50 mg/kg of BPA (1 mL) once a week (Group D).

Concentrations of serum luteinizing hormone (Endocrine Technologies, CA, USA), follicle stimulating hormone (Endocrine Technologies, CA, USA), and testosterone were measured by an ELISA plate reader (Versa Max, Molecular device, CA, USA) using the manufacturer's protocol of ELISA kits (Omega Diagnostics, Scotland, UK) for the hormone levels assay.

Following one month of treatment, rats were sacrificed and their testes were isolated. Testes were cut

into sections weighing about 30–50 mg and put in the protein extraction solution (Pro-prepTM) (Intron Biotechnology, Sungnam, Korea). Testes proteins were then homogenized by pestle and mortar. Seventy micrograms of testes protein were used for the western blot (the same method as in the *in vitro* western blot).

Terminal dUTP nick-end labelling (TUNEL) assay

Deparaffinized and rehydrated sections (5 μ m) were treated with 3% hydrogen peroxide for 10 min at room temperature to quench endogenous peroxidase, then treated with 20 μ g/mL proteinase K for 10 min at 37°C . Sections were then washed three times in $1\times$ phosphate buffered saline. Subsequent steps for terminal deoxynucleotide transferase-mediated TUNEL staining were carried out using the *in situ* Cell Death Detection, POD Kit (Roche, Mannheim, Germany) according to the supplier's instructions, except that the terminal transferase enzyme (TdT) was diluted 1:10 with sterile water. The sections obtained through the POD kit were incubated with the diluted TUNEL reaction mixture for 60 min at 37°C . The slides were then treated with 3% BSA blocking solution for 25 min at room temperature and incubated with the secondary anti fluorescein-POD-conjugated antibody for 30 min. After washing four times in PBST, diaminobenzidine chromogenic reagent was added to the sections. The sections were then counterstained with hematoxylin, dehydrated in a graded series of ethanol, and cleared in xylene. The positive cells were stained brown for observation with light microscopy²⁷.

Sperm cell counts in epididymides

Epididymides were cut in 30 mL saline at 37°C , and then incubated at 37°C for 30 min to get spermatozoa. Approximately 10 μ L of sperm suspension was placed on a haemocytometer for observation with light microscopy. Total sperm cell number was estimated using a haemocytometer. In addition, non-motile and motile sperm were counted. Sperm motility was expressed as a percentage of motile sperm of the total sperm count according to the method described by Kim *et al.*⁵².

Statistical analysis

The data is expressed as mean $\pm$ SD values. Statistical analyses were performed using the SigmaPlot 10.0 program (Systat Software Inc., Chicago, USA). Values were compared to the control using analysis of variance (ANOVA) followed by the unpaired Student's *t*-test. *P* values <0.05 were considered significant.

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Human and animal rights The experimental procedures followed the actual law of animal protection that was approved by the Animal Care Advisory Committee of Kyungseong University, Korea.

Conflict of Interest Seon Ok declares that he has no conflict of interest. Jae Seon Kang declares that he has no conflict of interest. Kang Min Kim declares that he has no conflict of interest.

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